



Problems

Year	Problem Title	Student Level	Source	Commentary	Student Papers
2017	Jet Lag	High School	IM ² C	No	No
2017	Optimizing the Passenger Throughput at an Airport Security Checkpoint	Undergraduate	ICM	No	Yes (5)
2017	Sustainable Cities Needed!	Undergraduate	ICM	No	Yes (5)
2017	Migration to Mars: Utopian Workforce of the 2100 Urban Society	Undergraduate	ICM	No	Yes (4)
2017	"Cooperate and navigate"	Undergraduate	MCM	No	Yes (4)
2017	Merge After Toll	Undergraduate	MCM	No	Yes (5)
2017	Managing The Zambezi River	Undergraduate	MCM	No	Yes (4)
2016	Swim, Bike, and Run	High School	HiMCM	Yes (1)	Yes (4)
2016	Shop and Ship	High School	HiMCM	Yes (1)	Yes (5)
2016	Record Insurance	High School	IM ² C	Yes (1)	Yes (3)
2016	A Hot Bath	Undergraduate	MCM	Yes (1)	Yes (6)
2016	Space Junk	Undergraduate	MCM	Yes (1)	Yes (3)
2016	The Goodgrant Challenge	Undergraduate	MCM	Yes (1)	Yes (4)
2016	Measuring the Evolution and Influence in Society's Information Networks	Undergraduate	ICM	Yes (1)	Yes (5)
2016	Are we heading towards a thirsty planet?	Undergraduate	ICM	Yes (1)	Yes (5)
2016	Modeling Refugee Immigration Policies	Undergraduate	ICM	Yes (1)	Yes (4)
2015	Preventing Road Rage	High School	HiMCM	Yes (1)	Yes (4)
2015	City Crime and Safety	High School	HiMCM	Yes (1)	Yes (4)
2015	Eradicating Ebola	Undergraduate	MCM	Yes (2)	Yes (5)
2015	Searching for a lost plane	Undergraduate	MCM	Yes (2)	Yes (5)
2015	Managing Human Capital in Organizations	Undergraduate	ICM	Yes (1)	Yes (5)
2015	Is it sustainable?	Undergraduate	ICM	Yes (1)	Yes (4)
2015	Movie Scheduling	High School	IM ² C	Yes (1)	Yes (4)
2014	Unloading Commuter Trains	High School	HiMCM	Yes (1)	Yes (5)
2014	The Next Plague?	High School	HiMCM	Yes (1)	Yes (4)
2014	The Keep-Right-Except-To-Pass Rule	Undergraduate	MCM	Yes (3)	Yes (6)
2014	College Coaching Legends	Undergraduate	MCM	Yes (3)	Yes (7)
2014	Using Networks to Measure Influence and Impact	Undergraduate	ICM	Yes (1)	Yes (6)
2013	Emergency Medical Response	High School	HiMCM	Yes (1)	Yes (5)
2013	Bank Service Problem	High School	HiMCM	Yes (1)	Yes (5)
2013	The Ultimate Brownie Pan	Undergraduate	MCM	Yes (1)	Yes (7)
2013	Water, Water, Everywhere	Undergraduate	MCM	Yes (2)	Yes (5)
2013	Network Modeling of Earth's Health	Undergraduate	ICM	Yes (1)	Yes (5)

2012	American Elk	High School	HiMCM	Yes (1)	Yes (3)
2012	How Much Gas Should I Buy This Week?	High School	HiMCM	Yes (1)	Yes (5)
2012	The Leaves of a Tree	Undergraduate	MCM	Yes (1)	Yes (5)
2012	Camping along the Big Long River	Undergraduate	MCM	Yes (3)	Yes (6)
2012	Modeling for Crime Busting	Undergraduate	ICM	Yes (1)	Yes (7)
2011	Space Shuttle Problem: No More Space Shuttles	High School	HiMCM	Yes (1)	Yes (4)
2011	Search and Find	High School	HiMCM	Yes (1)	Yes (4)
2011	Snowboard Course	Undergraduate	MCM	Yes (1)	Yes (4)
2011	Repeater Coordination	Undergraduate	MCM	Yes (1)	Yes (4)
2011	How environmentally and economically sound are electric vehicles? Is their widespread use feasible and practical?	Undergraduate	ICM	Yes (1)	Yes (6)
2010	Bicycle Club	High School	HiMCM	Yes (1)	Yes (3)
2010	Curbing City Violence	High School	HiMCM	Yes (1)	Yes (6)
2010	The Sweet Spot	Undergraduate	MCM	Yes (1)	Yes (4)
2010	Criminology	Undergraduate	MCM	Yes (2)	Yes (6)
2010	The Great Pacific Ocean Garbage Patch	Undergraduate	ICM	Yes (2)	Yes (4)
2009	Water, Water Everywhere	High School	HiMCM	Yes (1)	Yes (4)
2009	Tsunami ("Wipe Out!")	High School	HiMCM	Yes (1)	Yes (4)
2009	Designing a Traffic Circle	Undergraduate	MCM	Yes (1)	Yes (4)
2009	Energy and the Cell Phone	Undergraduate	MCM	Yes (2)	Yes (5)
2009	Creating Food Systems: Re-Balancing Human-Influenced Ecosystems	Undergraduate	ICM	Yes (2)	Yes (2)
2008	National Debt and National Crisis	High School	HiMCM	Yes (1)	Yes (3)
2008	Going Green	High School	HiMCM	Yes (1)	Yes (4)
2008	Take a Bath	Undergraduate	MCM	Yes (1)	Yes (4)
2008	Creating Sudoku Puzzles	Undergraduate	MCM	Yes (1)	Yes (5)
2008	Finding the Good in Health Care Systems	Undergraduate	ICM	Yes (1)	Yes (3)
2007	Smoke Alarms	High School	HiMCM	Yes (1)	Yes (1)
2007	Car Rentals	High School	HiMCM	Yes (1)	Yes (1)
2007	Gerrymandering	Undergraduate	MCM	No	Yes (5)
2007	The Airplane Seating Problem	Undergraduate	MCM	Yes (1)	Yes (8)
2007	Organ Transplant: The Kidney Exchange Problem	Undergraduate	ICM	Yes (3)	Yes (2)
2006	Inflation of the Parachute	High School	HiMCM	Yes (1)	Yes (1)
2006	A South Sea Island Resort	High School	HiMCM	Yes (1)	Yes (1)
2006	Positioning and Moving Sprinkler Systems for Irrigation	Undergraduate	MCM	Yes (1)	Yes (6)
2006	Wheel Chair Access at Airports	Undergraduate	MCM	Yes (1)	Yes (5)
2006	Trade-offs in the fight against HIV/AIDS	Undergraduate	ICM	Yes (2)	Yes (4)
2005	Modeling Ocean Bottom Topography	High School	HiMCM	Yes (1)	Yes (1)
2005	Gas Prices, Inventory, National Disasters, and the Mighty Dollar	High School	HiMCM	Yes (1)	Yes (1)
2005	Flood Planning	Undergraduate	MCM	Yes (1)	Yes (3)
2005	Tollbooths	Undergraduate	MCM	Yes (1)	Yes (7)
2005	Nonrenewable Resources	Undergraduate	ICM	Yes (2)	Yes (3)
2004	Motel Cleaning Problem	High School	HiMCM	Yes (1)	Yes (1)

2004	The Art Gallery Security System	High School	HiMCM	Yes (1)	Yes (1)
2004	Are Fingerprints Unique?	Undergraduate	MCM	Yes (3)	Yes (3)
2004	A Faster QuickPass System	Undergraduate	MCM	Yes (1)	Yes (4)
2004	To Be Secure or Not to Be?	Undergraduate	ICM	Yes (2)	Yes (4)
2003	What is it worth?	High School	HiMCM	Yes (1)	Yes (1)
2003	How fair are major league baseball parks to the players?	High School	HiMCM	Yes (1)	Yes (1)
2003	The Stunt Person	Undergraduate	MCM	Yes (1)	Yes (6)
2003	Gamma Knife Treatment Planning	Undergraduate	MCM	No	Yes (5)
2003	Aviation Baggage Screening Strategies: To Screen or Not to Screen, that is the Question	Undergraduate	ICM	Yes (2)	Yes (5)
2002	School Busing	High School	HiMCM	Yes (1)	Yes (1)
2002	The Falling Ladder	High School	HiMCM	Yes (1)	Yes (1)
2002	Wind and Waterspray	Undergraduate	MCM	Yes (1)	Yes (4)
2002	Airline Overbooking	Undergraduate	MCM	Yes (1)	Yes (6)
2002	If we SCRUB our land too much, we may lose the LIZARDS	Undergraduate	ICM	Yes (2)	Yes (2)
2001	Design of an Airline Terminal	High School	HiMCM	Yes (1)	Yes (1)
2001	Forest Service	High School	HiMCM	Yes (1)	Yes (1)
2001	Adolescent Pregnancy	High School	HiMCM	Yes (1)	Yes (1)
2001	Skyscrapers	High School	HiMCM	Yes (1)	Yes (1)
2001	The Bicycle Wheel Problem	Undergraduate	MCM	Yes (1)	Yes (3)
2001	The Hurricane Evacuation Problem	Undergraduate	MCM	Yes (1)	Yes (6)
2001	The Zebra Mussel Problem	Undergraduate	ICM	Yes (2)	Yes (3)
2000	Bank Robbers	High School	HiMCM	Yes (1)	Yes (1)
2000	Elections	High School	HiMCM	Yes (1)	Yes (1)
2000	Air Traffic Control	Undergraduate	MCM	No	No
2000	Radio Channel Assignments	Undergraduate	MCM	No	No
2000	Elephants: When is Enough, Enough?	Undergraduate	ICM	Yes (1)	Yes (3)
1999	Traffic Lights	High School	HiMCM	Yes (1)	Yes (1)
1999	Deep Impact	Undergraduate	MCM	No	No
1999	Unlawful Assembly	Undergraduate	MCM	No	No
1999	Ground Pollution	Undergraduate	ICM	Yes (2)	Yes (2)
1998	The Scanner Problem	Undergraduate	MCM	No	No
1998	The Grade Inflation Problem	Undergraduate	MCM	No	No
1997	The Velociraptor Problem	Undergraduate	MCM	No	No
1997	Mix Well For Fruitful Discussions	Undergraduate	MCM	No	No
1996	The Submarine Detection Problem	Undergraduate	MCM	Yes (2)	Yes (4)
1996	The Contest Judging Problem	Undergraduate	MCM	Yes (2)	Yes (4)
1995	The Single Helix	Undergraduate	MCM	Yes (3)	Yes (3)
1995	Aluacha Balaclava College	Undergraduate	MCM	Yes (1)	Yes (4)
1994	The Communications Network Problem	Undergraduate	MCM	No	No
1994	The Concrete Slab Problem	Undergraduate	MCM	No	No
1993	The Coal-Tipple Operations Problem	Undergraduate	MCM	No	No
1993	The Optimal Composting Problem	Undergraduate	MCM	No	No
1992	The Emergency Power-Restoration	Undergraduate	MCM	No	No

	<u>Problem</u>				
1992	<u>The Air-Traffic-Control Radar Problem</u>	Undergraduate	MCM	No	No
1991	<u>The Steiner Tree Problem</u>	Undergraduate	MCM	No	No
1991	<u>The Water Tank Problem</u>	Undergraduate	MCM	No	No
1990	<u>The Brain-Drug Problem</u>	Undergraduate	MCM	No	No
1990	<u>The Snowplow Problem</u>	Undergraduate	MCM	No	No
1989	<u>The Midge Classification Problem</u>	Undergraduate	MCM	No	No
1989	<u>The Aircraft Queueing Problem</u>	Undergraduate	MCM	No	No
1988	<u>The Drug Runner Problem</u>	Undergraduate	MCM	No	No
1988	<u>The Railroad Flatcar Problem</u>	Undergraduate	MCM	No	No
1987	<u>The Salt Storage Problem</u>	Undergraduate	MCM	No	No
1987	<u>The Parking Lot Problem</u>	Undergraduate	MCM	No	No
1986	<u>The Hydrographic Data Problem</u>	Undergraduate	MCM	Yes (1)	Yes (1)
1986	<u>The Emergency-Facilities Location Problem</u>	Undergraduate	MCM	Yes (1)	Yes (1)
1985	<u>The Strategic Reserve Problem</u>	Undergraduate	MCM	No	Yes (1)
1985	<u>The Animal Population Problem</u>	Undergraduate	MCM	No	Yes (1)